

Answer ALL questions. ALL working should be shown.

1. How do you describe the position of a rigid body?

[1 mark]

You will need:

- a reference frame (coordinate system)
- a frame rigidly attached to the rigid body B
- a vector from the origin of the reference frame to the origin of the frame which is rigidly attached to the rigid body B.

[1 mark]

2. Show that the rotation matrix ${}^A_B R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is orthonormal.

[4 marks]

We need to show that each column and each row has unit magnitude, and that each column and row is orthogonal (90 degrees) to other column and row.

[1 mark]

- Unit magnitudes:

- First column: Magnitude = $\sqrt{(\cos \theta)^2 + (\sin \theta)^2 + (0)^2} = 1$
- Second column: Magnitude = $\sqrt{(-\sin \theta)^2 + (\cos \theta)^2 + (0)^2} = 1$
- Third column: Magnitude = $\sqrt{(0)^2 + (0)^2 + (1)^2} = 1$
- First row: Magnitude = $\sqrt{(\cos \theta)^2 + (-\sin \theta)^2 + (0)^2} = 1$
- Second row: Magnitude = $\sqrt{(\sin \theta)^2 + (\cos \theta)^2 + (0)^2} = 1$
- Third row: Magnitude = $\sqrt{(0)^2 + (0)^2 + (1)^2} = 1$

[1.5 marks]

- Orthogonality:

- First & second column: $(\cos \theta)(-\sin \theta) + (\sin \theta)(\cos \theta) + (0)(0) = 0$
- First & third column: $(\cos \theta)(0) + (\sin \theta)(0) + (0)(1) = 0$
- Second & third column: $(-\sin \theta)(0) + (\cos \theta)(0) + (0)(1) = 0$
- First & second row: $(\cos \theta)(\sin \theta) + (-\sin \theta)(\cos \theta) + (0)(0) = 0$
- First & third row: $(\cos \theta)(0) + (-\sin \theta)(0) + (0)(1) = 0$
- Second & third row: $(\sin \theta)(0) + (\cos \theta)(0) + (0)(1) = 0$

[1.5 marks]

3. Frame $\{B\}$ is obtained by rotating $\{A\}$ along the z -axis by 45 deg. In frame $\{B\}$, the point P has a coordinate of $[2 \ 3 \ 1]^T$. What is the coordinate of point P in frame $\{A\}$? Keep your answer in fraction instead of decimal points.

[2.5 marks]

- First, get description of $\{B\}$ in $\{A\}$: ${}^A_B R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[1 mark]

- Then, calculate: ${}^A P = {}^A_B R \cdot {}^B P = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\frac{\sqrt{2}}{2} - 3\frac{\sqrt{2}}{2} + 0 \\ 2\frac{\sqrt{2}}{2} + 3\frac{\sqrt{2}}{2} + 0 \\ 0 + 0 + 1 \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ 5\frac{\sqrt{2}}{2} \\ 1 \end{bmatrix}$

[1.5 mark]

4. Given ${}^AP = {}^AR \cdot {}^BP + {}^AP_{BORG}$, write the equation using homogeneous transform.

[0.75 marks]

$$\begin{bmatrix} {}^AP \\ 1 \end{bmatrix} = \begin{bmatrix} {}^AR & {}^AP_{BORG} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^BP \\ 1 \end{bmatrix}$$

[0.75 marks]

5. Given ${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}$, derive the inverse of ${}^A_B T$. Note: you will need to show the derivation, not just give the final answer from your memory.

[2.5 marks]

- ${}^A P = {}^A_B T \cdot {}^B P$ gives ${}^A P$ in terms of ${}^B P \rightarrow {}^A P = {}^A_B R \cdot {}^B P + {}^A P_{BORG}$.
- If we can get ${}^B P$ in terms of ${}^A P$, then we would get the inverse of ${}^A_B T$.
- So, from the first line above:
 - ${}^A_B R \cdot {}^B P = {}^A P - {}^A P_{BORG}$ [0.5 marks]
 - ${}^B P = {}^A_B R^{-1} ({}^A P - {}^A P_{BORG}) = {}^A_B R^{-1} {}^A P - {}^A_B R^{-1} {}^A P_{BORG}$ [0.5 marks]
 - $\begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R^{-1} & -{}^A_B R^{-1} {}^A P_{BORG} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^A P \\ 1 \end{bmatrix}$ [1 marks]
- Thus the inverse of ${}^A_B T$ is:
 - ${}^A_B T^{-1} = \begin{bmatrix} {}^A_B R^{-1} & -{}^A_B R^{-1} {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}$
- But we also know that the inverse of rotation matrix is the same as its transpose, thus:
 - ${}^A_B T^{-1} = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}$ [0.5 marks]

6. Given ${}^A_B R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} cac\beta & cas\beta s\gamma - sac\gamma & cas\beta c\gamma + sas\gamma \\ sac\beta & sas\beta s\gamma + cac\gamma & sas\beta c\gamma - cas\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$. Calculate α , β and γ .

Specify also the condition when some answers cannot be obtained. (Note: $c\alpha$ is the shorthand notation for $\cos \alpha$, $s\alpha$ is the shorthand notation for $\sin \alpha$. Similar abbreviation for β and γ .)

[3 marks]

(Note: there can be different ways of solving the question)

- First, we notice that $-s\beta = r_{31}$ or $s\beta = -r_{31}$.
- Thus $c\beta = \pm\sqrt{1 - s^2\beta} = \pm\sqrt{1 - r_{31}^2}$.
- So $\beta = \text{atan2}(-r_{31}, \pm\sqrt{1 - r_{31}^2})$. [1 mark]
- Next, we notice that $c\beta s\gamma = r_{32}$ and $c\beta c\gamma = r_{33}$.
- Thus $s\gamma = \frac{r_{32}}{c\beta}$ and $c\gamma = \frac{r_{33}}{c\beta}$.
- So $\gamma = \text{atan2}\left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}\right)$ [1 mark]
- Finally, we notice that $cac\beta = r_{11}$ and $sac\beta = r_{21}$.
- Thus $c\alpha = \frac{r_{11}}{c\beta}$ and $s\alpha = \frac{r_{21}}{c\beta}$.
- So $\alpha = \text{atan2}\left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}\right)$ [0.5 mark]
- There is no solution when $c\beta = 0$, or $\beta = \pm 90^\circ$. [0.5 mark]

7. What is the advantage of using Euler parameters to represent orientation?

[0.5 marks]

There will always be solutions to $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4$.

[0.5 marks]

8. How many combinations are there for Fixed Angles representation of orientation? What are they?

[1 mark]

Twelve.

[0.5 marks]

- XYZ, YXZ, XZY, XZX,
- YXZ, YXY, YZX, YZY,
- ZXY, ZXZ, ZYX, ZYZ

[0.5 marks]

9. How many parameters are there in DH Parameters? And what are they (full names and symbols)?

[0.75 marks]

Four parameters. They are the link lengths a , link twists α , link offsets d and joint angles θ .

[0.75 marks]

10. In your own words, explain the general idea of solving the forward kinematics problem using systematic method of DH parameters.

[3 marks]

First we draw axes, and then we determine the mutual perpendiculars between axes. Frames will be attached at each joint, with z-axis pointing along the axis, and x-axis pointing along the mutual perpendiculars.

[1 mark]

With the frames fixed, we can then determine the DH parameters, followed by calculation of the transformation matrices from lower axis to higher axis. At the end, we calculate the compound transformation, and we will get the transformation matrix from base to the last joint.

[1 mark]

The end-effector position is fixed with respect to the last joint, and we can then calculate the end-effector position with respect to the base using mapping equation.

[1 mark]

11. Given the robot as shown in Fig 11.1, derive the DH parameters and summarize the results in a DH Table.

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.
- If there is any ambiguity in the diagonal direction, choose left + up.

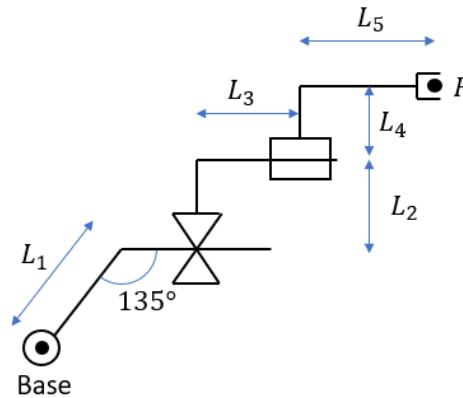
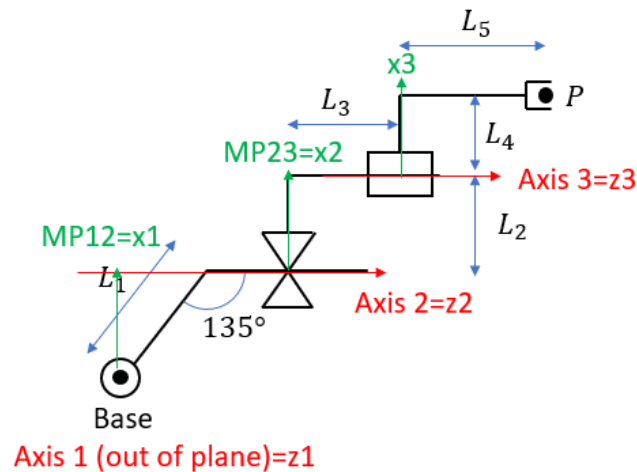


Fig. 11.1 Schematics of an RPR Robot

[7 marks]

Draw axes (red) and mutual perpendicular (green):



Frame attachment: z-axis along the axis, and x-axis along mutual perpendicular.

[2 mark]

Link lengths:

- $a_0 = \text{distance from } z_0 \text{ to } z_1 \text{ along } x_0 = 0.$
- $a_1 = \text{distance from } z_1 \text{ to } z_2 \text{ along } x_1 = L_1 \cos 45^\circ = \frac{\sqrt{2}}{2} L_1.$
- $a_2 = \text{distance from } z_2 \text{ to } z_3 \text{ along } x_2 = L_2.$

[1 mark]

Link twists:

- $\alpha_0 =$ angle from z_0 to z_1 about $x_0 = 0^\circ$.
- $\alpha_1 =$ angle from z_1 to z_2 about $x_1 = 90^\circ$.
- $\alpha_2 =$ angle from z_2 to z_3 about $x_2 = 0^\circ$.

[1 mark]

Link offsets:

- $d_1 =$ distance from x_0 to x_1 along $z_1 = 0$.
- $d_2 =$ distance from x_1 to x_2 along $z_2 =$ variable.
- $d_3 =$ distance from x_2 to x_3 along $z_3 = L_3$.

[1 mark]

Joint angles:

- $\theta_1 =$ angle from x_0 to x_1 about $z_1 =$ variable.
- $\theta_2 =$ angle from x_1 to x_2 about $z_2 = 0^\circ$.
- $\theta_3 =$ angle from x_2 to x_3 about $z_3 =$ variable.

[1 mark]

The DH table is therefore:

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0°	0	θ_1
2	$\frac{\sqrt{2}}{2} L_1$	90°	d_2	0°
3	L_2	0°	L_3	θ_3

[1 mark]

12. A DH table is as follows (Table 12.1):

Table 12.1 DH Table

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	3	30 degrees	0	θ_1
2	2	0 degrees	d_2	30 degrees
3	1	60 degrees	1	θ_3

The transformation matrix from frame $i - 1$ to frame i is given by:

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculate the transformation matrices 0_1T , 1_2T , 2_3T and 0_3T . (Note: $c\theta$ is the shorthand notation for $\cos \theta$, $s\theta$ is the shorthand notation for $\sin \theta$. Similar abbreviation for α .)

[23 marks]

For first row ($i = 1$), we have $s\alpha = \frac{1}{2}$, $c\alpha = \frac{\sqrt{3}}{2}$. Thus:

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 3 \\ \frac{\sqrt{3}}{2}s\theta_1 & \frac{\sqrt{3}}{2}c\theta_1 & -\frac{1}{2} & 0 \\ \frac{1}{2}s\theta_1 & \frac{1}{2}c\theta_1 & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[4 marks]

For second row ($i = 2$), we have $s\alpha = 0$, $c\alpha = 1$, $s\theta = \frac{1}{2}$, $c\theta = \frac{\sqrt{3}}{2}$. Thus:

$${}^1_2T = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 2 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[4 marks]

For third row ($i = 3$), we have $c\alpha = \frac{1}{2}$, $s\alpha = \frac{\sqrt{3}}{2}$. Thus:

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 1 \\ \frac{1}{2}s\theta_3 & \frac{1}{2}c\theta_3 & -\frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2}s\theta_3 & \frac{\sqrt{3}}{2}c\theta_3 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[4 marks]

$$\begin{aligned}
 {}^0_3T &= {}^0_1T \cdot {}^1_2T \cdot {}^2_3T \\
 &= \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 3 \\ \frac{\sqrt{3}}{2}s\theta_1 & \frac{\sqrt{3}}{2}c\theta_1 & -\frac{1}{2} & 0 \\ \frac{1}{2}s\theta_1 & \frac{1}{2}c\theta_1 & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 2 \\ 1 & \sqrt{3} & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 1 & d_2 \end{bmatrix} \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 1 \\ \frac{1}{2}s\theta_3 & \frac{1}{2}c\theta_3 & -\frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2}s\theta_3 & \frac{\sqrt{3}}{2}c\theta_3 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \left(\frac{\sqrt{3}}{2}c\theta_1 - \frac{1}{2}s\theta_1\right) & \left(-\frac{1}{2}c\theta_1 - \frac{\sqrt{3}}{2}s\theta_1\right) & 0 & (2c\theta_1 + 3) \\ \left(\frac{3}{4}s\theta_1 + \frac{\sqrt{3}}{4}c\theta_1\right) & \left(-\frac{\sqrt{3}}{4}s\theta_1 + \frac{3}{4}c\theta_1\right) & -\frac{1}{2} & \left(\sqrt{3}s\theta_1 - \frac{1}{2}d_2\right) \\ \left(\frac{\sqrt{3}}{4}s\theta_1 + \frac{1}{4}c\theta_1\right) & \left(-\frac{1}{4}s\theta_1 + \frac{\sqrt{3}}{4}c\theta_1\right) & \frac{\sqrt{3}}{2} & \left(s\theta_1 + \frac{\sqrt{3}}{2}d_2\right) \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 1 \\ \frac{1}{2}s\theta_3 & \frac{1}{2}c\theta_3 & -\frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2}s\theta_3 & \frac{\sqrt{3}}{2}c\theta_3 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2}c\theta_1c\theta_3 - \frac{1}{2}s\theta_1c\theta_3 \\ -\frac{1}{4}c\theta_1s\theta_3 - \frac{\sqrt{3}}{4}s\theta_1s\theta_3 \end{pmatrix} & \begin{pmatrix} -\frac{\sqrt{3}}{2}c\theta_1s\theta_3 + \frac{1}{2}s\theta_1s\theta_3 \\ -\frac{1}{4}c\theta_1c\theta_3 - \frac{\sqrt{3}}{4}s\theta_1c\theta_3 \end{pmatrix} & \left(\frac{\sqrt{3}}{4}c\theta_1 + \frac{3}{4}s\theta_1\right) & \begin{pmatrix} \frac{\sqrt{3}}{2}c\theta_1 - \frac{1}{2}s\theta_1 \\ +\frac{\sqrt{3}}{4}c\theta_1 + \frac{3}{4}s\theta_1 \\ +2c\theta_1 + 3 \end{pmatrix} \\ \begin{pmatrix} \frac{3}{4}s\theta_1c\theta_3 + \frac{\sqrt{3}}{4}c\theta_1c\theta_3 \\ -\frac{\sqrt{3}}{8}s\theta_1s\theta_3 + \frac{3}{8}c\theta_1s\theta_3 \\ -\frac{\sqrt{3}}{4}s\theta_3 \end{pmatrix} & \begin{pmatrix} -\frac{3}{4}s\theta_1s\theta_3 - \frac{\sqrt{3}}{4}c\theta_1s\theta_3 \\ -\frac{\sqrt{3}}{8}s\theta_1c\theta_3 + \frac{3}{8}c\theta_1c\theta_3 \\ -\frac{\sqrt{3}}{4}c\theta_3 \end{pmatrix} & \begin{pmatrix} \frac{3}{8}s\theta_1 - \frac{3\sqrt{3}}{8}c\theta_1 \\ -\frac{1}{4} \end{pmatrix} & \begin{pmatrix} \frac{3}{4}s\theta_1 + \frac{\sqrt{3}}{4}c\theta_1 \\ +\frac{3}{8}s\theta_1 - \frac{3\sqrt{3}}{8}c\theta_1 \\ -\frac{1}{4} \\ +\sqrt{3}s\theta_1 - \frac{1}{2}d_2 \end{pmatrix} \\ \begin{pmatrix} \frac{\sqrt{3}}{4}s\theta_1c\theta_3 + \frac{1}{4}c\theta_1c\theta_3 \\ -\frac{1}{8}s\theta_1s\theta_3 + \frac{\sqrt{3}}{8}c\theta_1s\theta_3 \\ +\frac{3}{4}s\theta_3 \end{pmatrix} & \begin{pmatrix} -\frac{\sqrt{3}}{4}s\theta_1s\theta_3 - \frac{1}{4}c\theta_1s\theta_3 \\ -\frac{1}{8}s\theta_1c\theta_3 + \frac{\sqrt{3}}{8}c\theta_1c\theta_3 \\ +\frac{3}{4}c\theta_3 \end{pmatrix} & \begin{pmatrix} \frac{\sqrt{3}}{8}s\theta_1 - \frac{3}{8}c\theta_1 \\ +\frac{\sqrt{3}}{4} \end{pmatrix} & \begin{pmatrix} \frac{\sqrt{3}}{4}s\theta_1 + \frac{1}{4}c\theta_1 \\ +\frac{\sqrt{3}}{8}s\theta_1 - \frac{3}{8}c\theta_1 \\ +\frac{\sqrt{3}}{4} \\ +s\theta_1 + \frac{\sqrt{3}}{2}d_2 \end{pmatrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

[Writing the matrices multiplication equation – 1 mark]

[Multiply 2 matrices – 5 marks]

[Multiply 3 matrices - 5 marks]

13. What are the two fundamental questions which need to be answered when solving inverse kinematics problems?

[0.5 marks]

Existence of solutions, uniqueness of solutions.

[0.5 marks]

14. When would an inverse kinematics problem have NO solution?

[0.5 marks]

When the target point lies outside of the manipulator's workspace.

[0.5 marks]

15. Sketch the workspace of the robot shown in Fig 15.1. Assume that the prismatic joint has a range of 0 m and 1 m, and that the revolute joints have no limits.

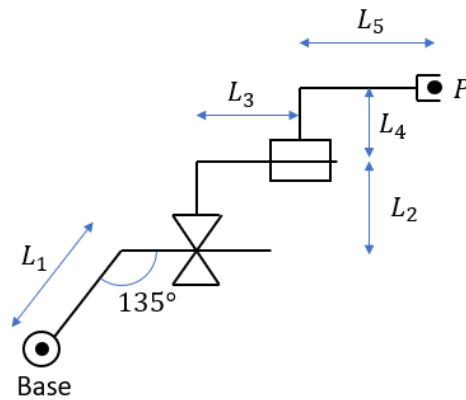
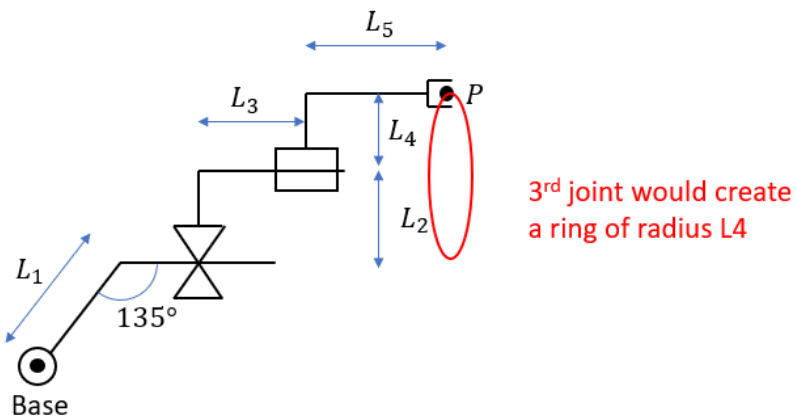
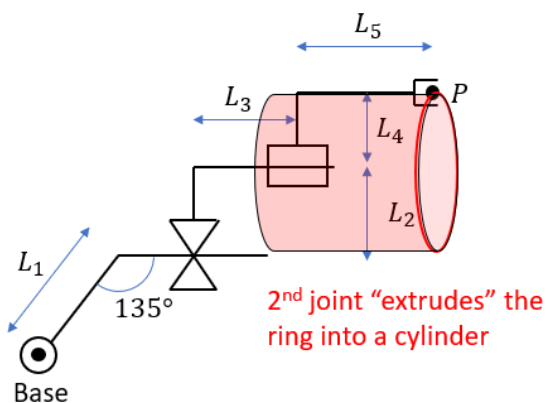


Fig 15.1 Schematics of an RPR Robot

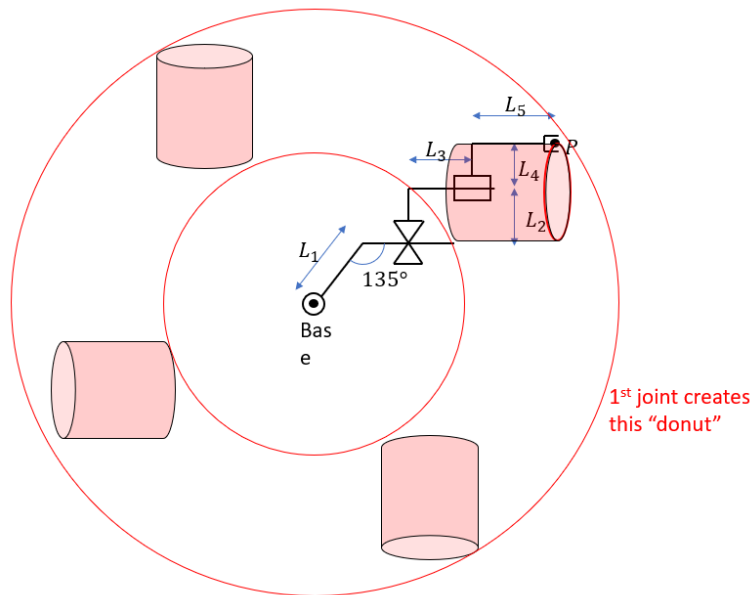
[3 marks]



[1 marks]



[1 marks]



[1 mark]

16. Given a robot as shown in Fig 16.1, derive the inverse kinematics of the robot if the end-effector is required to be at ${}^0P = [p_x \ p_y \ p_z]^T$ using the geometric approach.

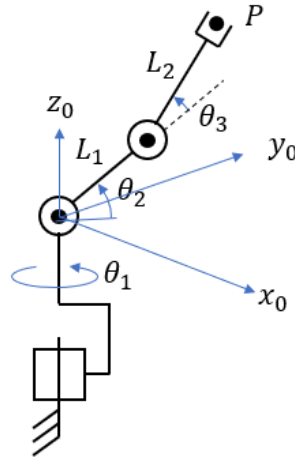
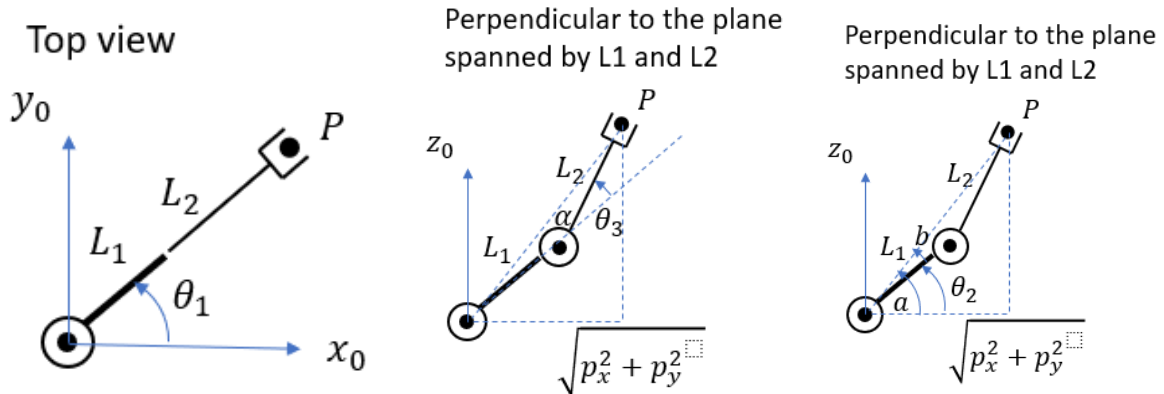


Fig 16.1 Schematics of an RRR Robot

[6 marks]



[1 mark for sketches]

From top view, the 2nd link and 3rd link form a straight line. It is easy to deduce that:

$$\theta_1 = \text{atan2}(y_0, x_0)$$

[1 mark]

Viewing the robot perpendicular to the plane spanned by 2nd link and 3rd link (middle picture), we see that:

$$\cos \alpha = \frac{L_1^2 + L_2^2 - (p_x^2 + p_y^2 + p_z^2)}{2L_1L_2}$$

Which gives:

$$\alpha = \cos^{-1} \left(\frac{L_1^2 + L_2^2 - (p_x^2 + p_y^2 + p_z^2)}{2L_1L_2} \right)$$

Therefore,

$$\theta_3 = 180^\circ - \alpha$$

[1 mark]

For the last picture, we have:

$$a = \text{atan2}\left(z_0, \sqrt{p_x^2 + p_y^2 + p_z^2}\right)$$

$$b = \cos^{-1}\left(\frac{L_1^2 + (p_x^2 + p_y^2 + p_z^2) - L_2^2}{2L_1\sqrt{p_x^2 + p_y^2 + p_z^2}}\right)$$

This gives:

$$\theta_2 = a - b$$

[2 marks]

There is another elbow up solution:

$$\theta_2' = a + b$$

$$\theta_3' = -\theta_3$$

[1 mark]

17. What does kinematic singularity mean? What is the mathematical condition for this to happen?
[1 mark]

Kinematic singularity means that the end-effector loses the ability to move in a direction or rotate about a direction.

[0.5 marks]

This happens mathematically when the Jacobian matrix has no inverse.

[0.5 marks]

18. The end-effector has the following position vector with respect to frame $\{0\}$:

$${}^0P = \begin{bmatrix} L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) + L_2 \cos \theta_2 \\ d_3 \\ L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) + d_3 \sin \theta_2 \end{bmatrix}$$

where L_i 's are constants, while θ_i 's and d_i 's are variables. Calculate the linear Jacobian matrix of the end-effector with respect to frame $\{0\}$. Does the end-effector have any singularities? If yes, please specify when singularities would happen.

[13 marks]

$$x = L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) + L_2 \cos \theta_2$$

$$y = d_3$$

$$z = L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) + d_3 \sin \theta_2$$

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial d_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial d_3} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial d_3} \end{bmatrix}$$

$$= \begin{bmatrix} L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) - L_1 \sin \theta_1 \sin(\theta_1 + \theta_2) & L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) - L_2 \sin \theta_2 & 0 \\ 0 & 0 & 1 \\ -L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) - L_1 \sin \theta_1 \cos(\theta_1 + \theta_2) & -L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) + d_3 \cos \theta_2 & \sin \theta_2 \end{bmatrix}$$

[4 marks]

To find singularity, we need to find the determinant of J and set it to zero.

$$\det(J) = \begin{vmatrix} L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) - L_1 \sin \theta_1 \sin(\theta_1 + \theta_2) & L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) - L_2 \sin \theta_2 \\ -L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) - L_1 \sin \theta_1 \cos(\theta_1 + \theta_2) & -L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) + d_3 \cos \theta_2 \end{vmatrix}$$

$$= -L_1^2 \cos^2 \theta_1 \cos(\theta_1 + \theta_2) \sin(\theta_1 + \theta_2) + L_1 d_3 \cos \theta_1 \cos(\theta_1 + \theta_2) \cos \theta_2$$

$$+ L_1^2 \sin \theta_1 \cos \theta_1 \sin^2(\theta_1 + \theta_2) - L_1 d_3 \sin \theta_1 \sin(\theta_1 + \theta_2) \cos \theta_2$$

$$+ L_1^2 \cos^2 \theta_1 \cos(\theta_1 + \theta_2) \sin(\theta_1 + \theta_2) - L_1 L_2 \cos \theta_1 \sin(\theta_1 + \theta_2) \sin \theta_2$$

$$+ L_1^2 \sin \theta_1 \cos \theta_1 \cos^2(\theta_1 + \theta_2) - L_1 L_2 \sin \theta_1 \sin \theta_2 \cos(\theta_1 + \theta_2)$$

$$= L_1 d_3 \cos \theta_1 \cos(\theta_1 + \theta_2) \cos \theta_2$$

$$+ L_1^2 \sin \theta_1 \cos \theta_1 - L_1 d_3 \sin \theta_1 \sin(\theta_1 + \theta_2) \cos \theta_2$$

$$- L_1 L_2 \cos \theta_1 \sin(\theta_1 + \theta_2) \sin \theta_2$$

$$- L_1 L_2 \sin \theta_1 \sin \theta_2 \cos(\theta_1 + \theta_2)$$

$$= L_1 d_3 \cos \theta_2 (\cos \theta_1 \cos(\theta_1 + \theta_2) - \sin \theta_1 \sin(\theta_1 + \theta_2)) + L_1^2 \sin \theta_1 \cos \theta_1$$

$$- L_1 L_2 \sin \theta_2 (\cos \theta_1 \sin(\theta_1 + \theta_2) + \sin \theta_1 \cos(\theta_1 + \theta_2))$$

$$= L_1 d_3 \cos \theta_2 \cos(2\theta_1 + \theta_2) + L_1^2 \sin \theta_1 \cos \theta_1 - L_1 L_2 \sin \theta_2 \sin(2\theta_1 + \theta_2)$$

[6 marks]

When $d_3 = L_2$, we have:

$$\begin{aligned}\det(J) &= L_1 L_2 \cos \theta_2 \cos(2\theta_1 + \theta_2) + L_1^2 \sin \theta_1 \cos \theta_1 - L_1 L_2 \sin \theta_2 \sin(2\theta_1 + \theta_2) \\ &= L_1 L_2 \cos(2\theta_1 + 2\theta_2) + L_1^2 \sin \theta_1 \cos \theta_1\end{aligned}$$

Possible singularities when:

- $\theta_1 = 0^\circ, \theta_2 = 45^\circ, d_3 = L_2$
- $\theta_1 = 90^\circ, \theta_2 = 45^\circ, d_3 = L_2$

[3 marks]

19. What are the disadvantages of Cartesian space schemes in trajectory planning?

[0.75 marks]

Computationally expensive as inverse kinematics has to be solved. Prone to problems relating to workspace and singularities.

[0.75 marks]

20. Explain the procedure to calculate the parameters a_0, a_1, a_2, a_3 for a cubic polynomial $u(t) = a_0 + a_1t + a_2t^2 + a_3t^3$, where starting position is u_0 and final position is u_f . The time to reach final position is t_f .

[5.5 marks]

- Set $u(t = 0) = a_0 = u_0$ and we will get a_0 . [1 mark]
- Differentiate the cubic polynomial to get the velocity equation: $\dot{u}(t) = a_1 + 2a_2t + 3a_3t^2$.
- Set $\dot{u}(t = 0) = a_1 = 0$ and we will get a_1 . [1 mark]
- To get a_2 and a_3 we will need to solve the following simultaneous equation:

$$\circ u(t = t_f) = a_0 + a_1t_f + a_2t_f^2 + a_3t_f^3 = u_f$$

$$\circ \dot{u}(t = t_f) = a_1 + 2a_2t_f + 3a_3t_f^2 = 0$$

[1.5 mark]

- This gives:

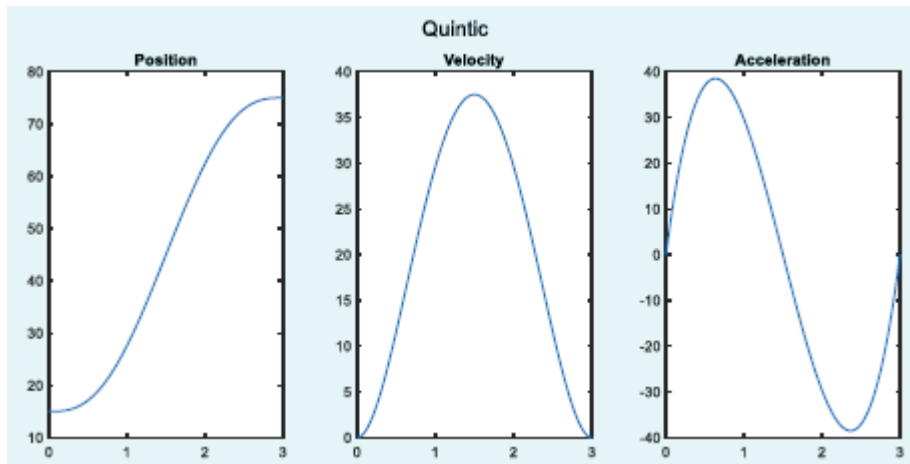
$$\circ \begin{bmatrix} t_f^2 & t_f^3 \\ 2t_f & 3t_f^2 \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} u_f - u_0 \\ 0 \end{bmatrix}$$

$$\circ \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{t_f^2}(u_f - u_0) \\ -\frac{2}{t_f^3}(u_f - u_0) \end{bmatrix}$$

[2 marks]

21. Sketch the position, velocity and acceleration profiles for a quintic polynomial. Assume u_0 is less than u_f .

[0.75 marks]



[0.75 marks]

22. Explain the general idea of calculating the dynamic equation of robots using the Newton Euler Formula.

[2.5 marks]

- Propagate velocity and acceleration from lower frame to upper frame.
- Propagate the acceleration of each frame to the centre of mass of each link.
- With acceleration of centre of mass, and the mass of link, we can calculate the force on the link via $F=ma$.
- Propagate joint forces (external force, motor torques) from higher frame to lower frame.
- The force on link, F , can then be related to the joint forces, and these will give dynamic equations.

[2.5 marks]

23. The dynamic equation for a 2-link robot is as follows:

$$\begin{aligned}\tau_1 &= m_2 L_2^2 (\ddot{\theta}_1 + \ddot{\theta}_2) + m_2 L_1 L_2 c_2 (2\ddot{\theta}_1 + \ddot{\theta}_2) + (m_1 + m_2) L_1^2 \ddot{\theta}_1^2 - m_2 L_1 L_2 s_2 \dot{\theta}_2 \\ &\quad - 2m_2 L_1 L_2 s_2 \dot{\theta}_1 \dot{\theta}_2 + m_2 g L_2 c_{12} + (m_1 + m_2) g L_1 c_1 \\ \tau_2 &= m_2 L_1 L_2 s_2 \dot{\theta}_1^2 + m_2 L_1 L_2 s_2 \ddot{\theta}_1 + m_2 g L_2 c_{12} + m_2 L_2^2 (\ddot{\theta}_1 + \ddot{\theta}_2)\end{aligned}$$

Are there any errors in the equation? If yes, please list them down. (Note: c_i is the shorthand notation for $\cos \theta_i$, s_i is the shorthand notation for $\sin \theta_i$, c_{ij} is the shorthand notation for $\cos(\theta_i + \theta_j)$, s_{ij} is the shorthand notation for $\sin(\theta_i + \theta_j)$)

[3 marks]

- In τ_1 equation, $(m_1 + m_2) L_1^2 \ddot{\theta}_1^2$ is wrong. $\ddot{\theta}_1^2$ should not have squares.

[1 mark]

- In τ_1 equation, $m_2 L_1 L_2 s_2 \dot{\theta}_2$ is wrong. $\dot{\theta}_2$ should be squared.

[1 mark]

- The M-matrix is not symmetrical.

[1 mark]

24. What is the equation for Coulomb friction?

[0.5 marks]

$$\tau_{\text{friction}} = c \cdot \text{sgn}(\dot{q})$$

[0.5 marks]

25. Given a rectangular block with length l , width w and height h as shown in Fig 25.1, calculate the mass moment of inertia I_{xx} . Assume the volume is V , the density ρ is uniform and the mass is m .

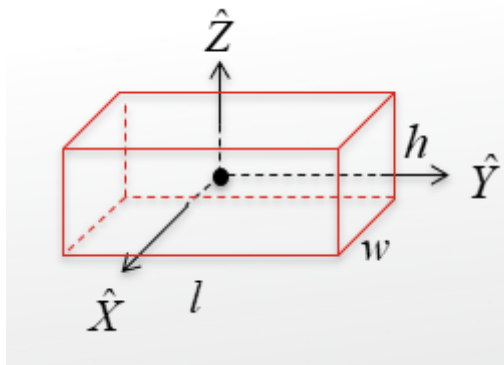


Fig. 25.1 Rectangular Block

(Hint: the formula for I_{xx} is $I_{xx} = \iiint_V (y^2 + z^2) \rho dV$.)

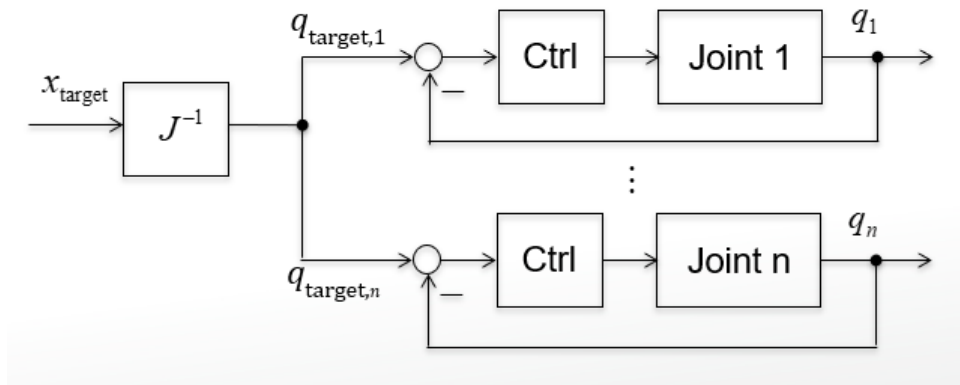
[5 marks]

$$\begin{aligned}
 I_{xx} &= \iiint_V (y^2 + z^2) \rho dV = \int_{-h/2}^{h/2} \int_{-l/2}^{l/2} \int_{-w/2}^{w/2} (y^2 + z^2) \rho dx \cdot dy \cdot dz \\
 &= \int_{-h/2}^{h/2} \int_{-l/2}^{l/2} (y^2 + z^2) w \rho \cdot dy \cdot dz = \int_{-h/2}^{h/2} \left(\frac{l^3}{12} + z^2 l \right) w \rho \cdot dz \\
 &= \left(\frac{l^3 h}{12} + \frac{h^3 l}{12} \right) w \rho = \left(\frac{l^2}{12} + \frac{h^2}{12} \right) \underbrace{h l w \rho}_m = \frac{m}{12} (l^2 + h^2)
 \end{aligned}$$

[5 marks]

26. Sketch a robot control block diagram, where the reference trajectory is given in Cartesian space, and the joints are controlled individually.

[1.5 marks]



[1.5 marks]

27. Determine the motion of a mass-spring-damper system if the parameter values are $m = 3, b = 2, k = 4$, and the mass (initially at rest) is released from position $x = 2$.

[5 marks]

The system is:

$$m\ddot{x} + b\dot{x} + kx = 0$$

And with values:

$$3\ddot{x} + 2\dot{x} + 4x = 0$$

The characteristic equation is:

$$3\lambda^2 + 2\lambda + 4 = 0$$

The roots of the equation are:

$$\lambda_1 = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 3 \cdot 4}}{2 \cdot 3} = \frac{-2 \pm \sqrt{-44}}{6} = -\frac{1}{3} \pm j \frac{\sqrt{44}}{6} = -\frac{1}{3} \pm j \frac{\sqrt{11}}{3}$$

The general solution for x is therefore:

$$x(t) = c_1 e^{-\frac{1}{3}t} \cos\left(\frac{\sqrt{11}}{3}t\right) + c_2 e^{-\frac{1}{3}t} \sin\left(\frac{\sqrt{11}}{3}t\right)$$

[2 marks]

The velocity equation is:

$$\dot{x}(t) = -\frac{1}{3}c_1 e^{-\frac{1}{3}t} \cos\left(\frac{\sqrt{11}}{3}t\right) - c_1 e^{-\frac{1}{3}t} \sin\left(\frac{\sqrt{11}}{3}t\right) - \frac{1}{3}c_2 e^{-\frac{1}{3}t} \sin\left(\frac{\sqrt{11}}{3}t\right) + c_2 e^{-\frac{1}{3}t} \cos\left(\frac{\sqrt{11}}{3}t\right)$$

To calculate c_1 and c_2 , put in boundary conditions:

$$x(t = 0) = c_1 = 2$$

$$\dot{x}(t = 0) = -\frac{1}{3}c_1 + c_2 = 0$$

$$c_2 = \frac{1}{3}c_1 = \frac{2}{3}$$

So the equation is:

$$x(t) = 2e^{-\frac{1}{3}t} \cos\left(\frac{\sqrt{11}}{3}t\right) + \frac{2}{3}e^{-\frac{1}{3}t} \sin\left(\frac{\sqrt{11}}{3}t\right)$$

[3 marks]

28. In what condition would a mass-spring-damper system experience an overdamped response?
[0.75 marks]

$b^2 > 4mk$. If students cannot remember this equation, they can also say if damping is high in comparison to mass and stiffness. If they say damping is high only (without referring to mass and stiffness), they will get zero marks because the question already says “overdamped”.

[0.75 marks]

29. What happens to the robot's dynamic behaviour if we use linear control to control each joint independently?

[0.75 marks]

There will be undesirable results, e.g. the damping will not be uniform throughout the workspace.

[0.75 marks]

30. In your own words, explain why trajectory following control leads to x tracking x_d .

[1 mark]

If we define tracking error as $e = x_d - x$

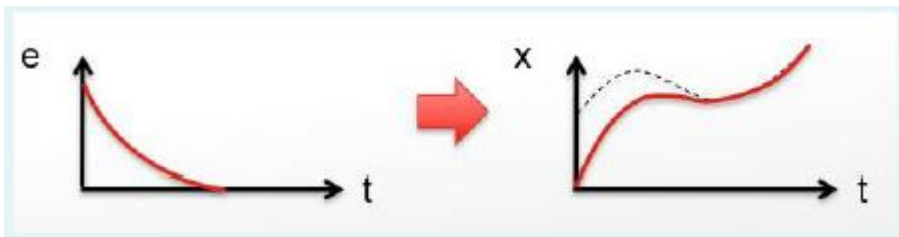
The closed loop equation then becomes:

$$\ddot{e} + k_v \dot{e} + k_p e = 0$$

e has the same differential equation as discussed earlier!

[0.5 marks]

Since e decreases to zero, x will track x_d eventually.



[0.5 marks]